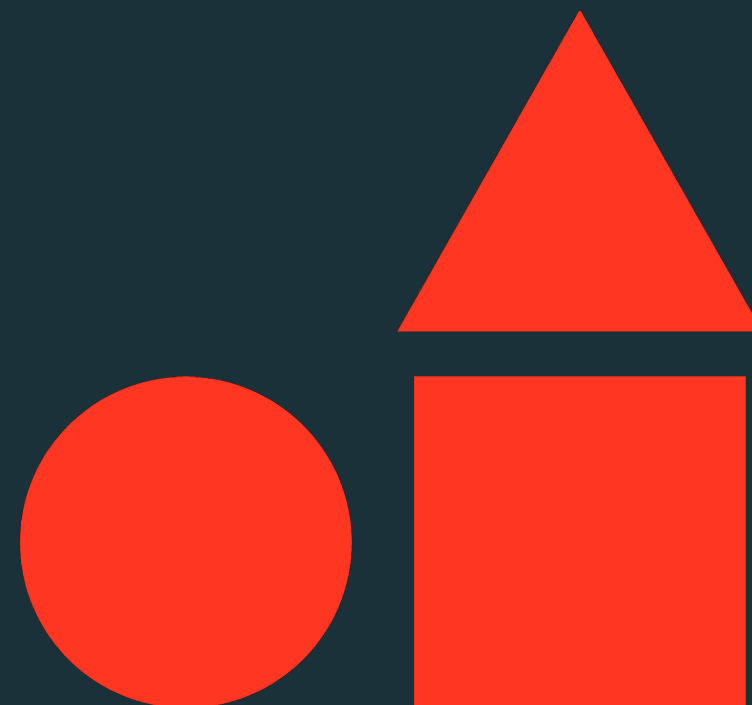


Northbay Energy Gauge Reader

AI visual inspection for
robot rounds

Starting with every pressure gauge | October 2026



Where the value comes from

Illustrative site: 500 gauges, photographed twice a day

~4,900 hours a year

- 365,000 readings a year, each transcribed or re-checked by hand: ~6,100 hours at 1 minute each¹
- With AI, people check only the uncertain ones (target: 1 in 5)
- About 3 people's time, freed for maintenance

~\$15k a year AI cost

- 365,000 photos × ~\$0.04 per photo at list price²
- vs. ~\$180k a year at a \$0.50 per-image vendor fee¹
- No model to train, host or maintain

Earlier warnings

- Excursions flagged the same shift, not at the next manual round
- Unplanned downtime costs a large plant ~\$0.8M an hour³
- Avoiding a single hour is worth ~50× the annual AI cost

¹ Assumptions to replace with Northbay's numbers: 1 min per reading, \$0.50 per image vendor fee. ² Measured tokens per photo × GPT-5.5 list price. ³ Siemens, The True Cost of Downtime 2024.



Value for both sides of the business

Each lever mapped to the numbers each leader is measured on

Executive sponsor — VP Operations / HSE

- **Measured on: recordable injury rate (TRIR), process-safety events, plant availability, operating cost**
- Safety: ~2,200 operator hours a year kept out of process areas as robot rounds produce trusted readings
- Availability: same-shift excursion warnings; one avoided hour of unplanned downtime ≈ \$0.8M
- Operating cost: ~\$15k a year on the platform vs. ~\$180k in per-image vendor fees

Domain owner — Reliability / Maintenance lead

- **Measured on: unplanned downtime hours, mean time between failures (MTBF), maintenance backlog**
- Downtime and MTBF: excursions caught between rounds, with per-gauge trends to spot degradation early
- Backlog: one review queue sorted by risk; unexpectedly high readings become work orders
- Audit: every correction recorded (who, when, why) and answerable in Genie



Why Northbay needs this

Robots photograph every gauge — trusted readings are the bottleneck

Analog gauges are here to stay

- The vast majority of plant gauges are still analog¹
- Swapping each for a transmitter means hot work, cabling and hazardous-area certification
- So the reading still has to come from a person, or a photo

A photo is not a reading

- Robots already capture photos at scale: 120,000+ inspections a month for one vendor alone³
- Glare, dirt, rain and steep angles defeat simple vision models¹
- Third-party readings still get re-checked by hand, outside the data platform

Late readings cost money

- An out-of-range pressure found at the next manual round is a missed early warning
- Unplanned downtime costs the world's 500 largest companies ~\$1.4tn a year, 11% of revenue²

¹ Reitsma et al, "Under pressure: learning-based analog gauge reading in the wild", ETH Zurich, ICRA 2024. ² Siemens, The True Cost of Downtime 2024. ³ ANYbotics, anybotics.com (vendor-reported).



What we hear from energy operators

Patterns from robot inspection programmes, and our answer to each

What operators tell us	How Northbay's Gauge Reader answers it
<i>"Accuracy comes first — a wrong reading is worse than a late one."</i>	Every reading carries a confidence score; low-confidence, unreadable and unexpectedly high readings always go to a person
<i>"We pay a vendor per image, and still check its readings by hand."</i>	Readings run inside the platform at a known cost per reading (~\$0.04), with every correction audited
<i>"Photos are taken in fog, rain and glare, on dirty dials."</i>	The model reports image issues (glare, blur, angle, dirt) so reviewers see why a reading was flagged
<i>"We want to own the capability, not rent a black box."</i>	A governed foundation model in Unity Catalog: no training data, swap models through Unity AI Gateway
<i>"Next: spot rust and leaks, and trends across rounds."</i>	The same pipeline extends to condition findings and per-gauge trends, feeding maintenance work orders



One platform for visual inspection

Gauge readings are the first use case — not the last

Beyond gauges

- The same photos also show leaks, corrosion and rust, breakages, missing guards and valve positions
- A new inspection type is a new prompt and output fields, not a new project
- Same review queue, same governance, same Genie

Open, not a black box

- Foundation vision models instead of hand-built computer vision that needs tuning per dial, angle and light
- No third-party vendor or per-image fee; prompts, data and results stay in your lakehouse
- Swap models through Unity AI Gateway

Specialise when it pays

- The model is an example: start with a foundation model, no training needed
- Every reviewer correction becomes a labelled example
- Fine-tune specialist models (gauge reading, rust detection) and evaluate them before switching

Any cloud, any robot

- Runs on Databricks on AWS, Azure and Google Cloud, deployed as one bundle¹
- Works with photos from any robot, drone or handheld camera
- No lock-in to a robot or vision vendor

¹ Lakebase is in Beta on Google Cloud; where Lakehouse Sync (Lakebase → Delta) is not yet available, a scheduled MERGE returns review decisions to Delta (DECISIONS D4).



Beyond classic vision and vendors

Foundation models on an open platform change the economics

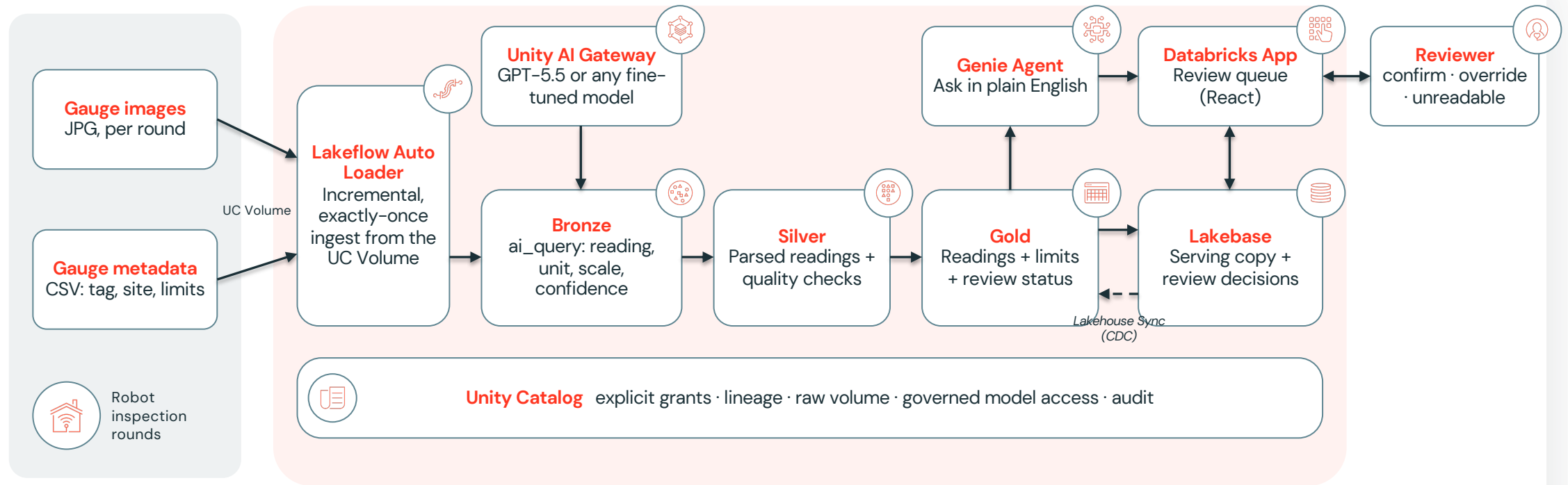
	Bespoke computer vision	Third-party vendor	Foundation models on Databricks
New inspection type	New detection, segmentation and OCR models to build	Wait for the vendor's roadmap	New prompt and output fields; test it the same day
Labelled data to start	Labelled images for every gauge type and defect	None, but it's the vendor's model	None — reviewer corrections build a labelled set as you go
Hand-tuning	Per dial design, angle, glare and lighting	Opaque; results still re-checked by hand	Confidence score routes uncertain readings to a person
Cost model	GPU training plus a team to maintain it	Fee per image	Pay per token: ~\$0.04 per gauge reading²
Data and governance	A separate stack to secure	Results arrive outside your platform	Unity Catalog: grants, lineage, audit
Improves over time	Only when someone retrains it	On the vendor's terms	Swap or fine-tune models through Unity AI Gateway

² Build run: GPT-5.5 via Unity AI Gateway at list price (evidence/07_gateway.md). Bespoke pipeline steps as described in Reitsma et al., ETH Zurich, ICRA 2024.



Architecture

One governed journey: robot photo → trusted reading → answer



Human in the loop confidence < 70%, unreadable or unexpectedly high → a reviewer; decisions appended, never overwritten

Unity Catalog analysts see gold only; the app reads photos read-only; Genie runs as the signed-in user

Unity AI Gateway every model call attributed and costed (~\$0.04 per reading); swap models without pipeline changes



Decisions and trade-offs

Why this approach and not another

- Foundation vision model via ai_query, not a custom CNN: no labelled data or training; confidence-based review covers the gaps
- Model called inside the streaming table: each image read and billed once; reviewed readings never change on refresh
- Lakebase for the review step: low-latency transactional writes; decisions return to Delta via Lakehouse Sync
- Snapshot sync for serving: gold is a materialized view; at ~120 images/day a snapshot takes seconds
- “Unexpectedly high” = above the gauge's operating limit (default 75% of full scale), so Genie never invents thresholds
- GPT-5.5 chosen after testing 15 vision models: best accuracy among batch-capable models, with a confidence score that separates right from wrong



Roadmap and pilot

Each phase reuses the pipeline, review app, governance and Genie

- 1 · Gauge readings: pilot at one site; unexpectedly high readings raise maintenance work orders
 - 2 · Condition findings: leaks, rust and breakages from the same photos, in the same review queue
 - 3 · Specialist models: fine-tune on reviewer-labelled photos; evaluate before switching
 - Pilot set-up: connect the robots' image export, load gauge tags and limits from the asset register, promote dev → prod with bundles
- Pilot success criteria
 - One site, 8 weeks, ~500 gauges
 - At most 1 in 5 readings sent to review
 - $\geq 95\%$ of auto-accepted readings within $\pm 2\%$ of full scale (weekly spot check)
 - ~4,900 hours a year of reading and re-checking removed (run rate)
 - Every reading traceable from photo to answer



**Every photo read, every finding
governed**



Appendix A · Pipeline run

Evidence: Lakeflow event log — 30 photos read once

Pipeline update (serverless, triggered)

```
update 177c2109-d3f2-49aa-846d-8b434c3b6861    COMPLETED    full_refresh=False
started 2026-10-06 12:49:11 UTC    finished 12:51:52 UTC    duration 161 s
```

```
flow                type                output_rows
bronze_gauge_readings_ai    streaming table    30    <- Auto Loader + ai_query: one model call per photo
bronze_gauge_metadata    streaming table    30
silver_gauge_readings    streaming table    30
quarantine_vlm_errors    streaming table    0    <- no failed model calls
gold_gauge_readings_final    materialized view    30
Later updates incl. the refresh job (cause JOB_TASK): bronze_gauge_readings_ai = 0 rows each time -> no photo re-read
```

Data-quality expectations (event log)

expectation	passed	failed
vlm_call_succeeded	30	0
response_parsed	30	0
reading_within_scale	30	0
unit_present	29	1 (warn, routed to review)
gauge_id_present	30	0

Unity AI Gateway (system.ai_gateway.usage)

service	system.ai.gpt-5-5	source AI_QUERY
requests	33 (30 photos + 3 test calls)	
errors	0	
tokens	65,063 in / 31,717 out	
cost	\$1.2688 at list price	
	= \$0.0423 per photo	



Appendix B · Governance and serving

Evidence: Unity Catalog grants and Lakebase query results

Unity Catalog: SHOW GRANTS and system.access.table_lineage

```
gold_gauge_readings_final    account users          SELECT      (analysts: gold only)
raw (volume)                 app service principal READ VOLUME  (photos read-only)
bronze_gauge_readings_ai     no grants to analysts or the app
system.ai model services     account users          EXECUTE
lineage: files -> bronze -> silver -> gold; lb_reading_overrides_history (Lakebase decisions) -> gold
```

Lakebase: query on pressure_gauge.gauge_readings_serving (what the app reads)

```
rows | needs_review | unexpectedly_high | human_corrected
30 | 9 | 4 | 1
```

image_id	gauge_id	site	value	unit	confidence	reading_status	review_status
gauge_004.jpg	PI-1101	Northbay Refinery		unknown	0.00	unreadable	pending_review
gauge_003.jpg	PI-6102	Ridgeway Terminal	190	psi	0.42	normal	pending_review
gauge_030.jpg	PI-4108	Northbay Refinery	620	kPa	0.62	high	pending_review
gauge_013.jpg	PI-3103	Northbay Refinery	187	psi	0.86	high	pending_review



Appendix C · Write-back and the app

Evidence: one human correction end to end; the deployed app's live API

Correction round trip: app -> Lakebase -> Delta -> gold -> Lakebase

```
1 review.reading_overrides (Lakebase)    id 2 | gauge_009.jpg | override | human_value 1 | 2026-10-06 13:28:01 UTC
   reason: "Read the wrong scale: needle tip is at ~1.0 bar; the AI read the counterweight end"
2 Lakehouse Sync -> Delta history        insert | _pg_lsn 30799832 | 2026-10-06T13:28:01.399
3 Pipeline (job run) -> gold             PI-3102 | ai_value 5.0 | final_value 1.0 bar | overridden | is_human_corrected true
4 Snapshot sync -> serving table         gauge_009.jpg | ai_value 5 | final_value 1 | overridden
```

Deployed Databricks App: live API responses

```
GET /api/readings      -> HTTP 200    30 readings · reviewConfidenceThreshold 0.7
  { "gauge_id": "PI-3102", "ai_value": 5, "final_value": 1, "unit": "bar", "vlm_confidence": 0.86,
    "review_status": "overridden", "override_reason": "Read the wrong scale: needle tip is at ~1.0 bar ..." }
GET /api/model-usage   -> HTTP 200    { "requests": "33", "images_read": "30", "errors": "0", "est_cost_usd": 1.2688 }
GET /api/review-stats  -> HTTP 200    { "decisions": 1, "images_reviewed": 1 }
```



Appendix D • Genie Agent

Evidence: Genie's generated SQL and answers

Q: Are any pressure readings unusually high?

```
SELECT captured_at, image_id, gauge_id, site, unit_area, final_value, unit, normal_max,  
       (final_value - normal_max) AS above_normal_max  
FROM serverless_stable_kx6lwb_catalog.pressure_gauge.gold_gauge_readings_final  
WHERE reading_status = 'high' AND final_value IS NOT NULL AND normal_max IS NOT NULL  
ORDER BY above_normal_max DESC
```

PI-3103		Steam & Utilities		187.0 psi		max 175.0		+12.0		PI-4108		Instrument Air		620.0 kPa		max 615.0		+5.0
PI-3104		Steam & Utilities		68.0 psi		max 60.0		+8.0		PI-2101		Hydrotreater		20.0 bar		max 18.75		+1.25

Answer: "Yes - 4 pressure readings are above their normal maximums ... the largest excursion is PI-3103."

Q: Which image readings have needed human intervention?

```
SELECT image_id, gauge_id, site, ai_value, human_value, unit, override_reason, reviewed_by, reviewed_at  
FROM ...gold_gauge_readings_final WHERE is_human_corrected = true ORDER BY reviewed_at DESC
```

Answer: "1 image reading needed human intervention: gauge_009.jpg for gauge PI-3102 at Northbay Refinery.
The AI reading was 5.0 bar, the human-corrected reading was 1.0 bar ..."

